

7.2

Trigonometric Integral

Our goals are:

In this section, we will learn:

How to use trigonometric identities to integrate certain combinations of trigonometric functions.

Imprtant rules:

$$\int \sin u \, du = -\cos u + C$$

$$\int \cos u \, du = \sin u + C$$

$$\int \sec^2 u \, du = \tan u + c$$

$$\int \csc^2 u \, du = -\cot u + C$$

$$\int \sec u \tan u \, du = \sec u + C$$

$$\int \csc u \cot u \, du = -\csc u + C$$

Example:

- Evaluate $\int \cos^3 x \, dx$

Example:

- Find $\int \sin^5 x \cos^2 x \, dx$

Note that, if the powers of both sine and cosine are odd, either (A) or (B) can be used.

Remark:

In the preceding examples, an odd power of sine or cosine enabled us to separate a single factor and convert the remaining even power.

If the integrand contains even powers of both sine and cosine, this strategy fails.

In that case, we can take advantage of the following half-angle identities:

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

Example:

- Evaluate

$$\int_0^{\pi} \sin^2 x \, dx$$

Example:

Find $\int \sin^4 x \, dx$

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$$\int \sin^5 x dx = \int (\sin x)(\sin^2 x)^2 \, dx = \int (\sin x)(1-\cos^2 x)^2 \, dx$$

$$= \int -(1-u^2)^2 \, du$$

$$= \int (-u^4 + 2u^2 - 1) du$$

$$= \frac{-u^5}{5} + \frac{2u^3}{3} - u + C$$

$$= \frac{-\cos^5 x}{5} + \frac{2\cos^3 x}{3} - \cos x + C$$

- Find $\int \tan^2 x \sec^2 x \, dx$

- **To integrate $\int \tan^n x \sec^m x \, dx$ *where m is even***
- ***Save* $\sec^2 x$ *factor, and use* $\tan^2 x + 1 = \sec^2 x$**
- ***Use u=tanx***

Example

- Find $\int \tan^5 x \sec^3 x \, dx$

To integrate $\int \tan^n x \sec^m x \, dx$ *where n is odd*
Save tan sec factor, and use $\tan^2 x + 1 = \sec^2 x$
Use u=secx



Example

Find

- $\int \tan^3 x \, dx$

Remark:

- **If an even power of tangent appears with an odd power of secant, it is helpful to express the integrand completely in terms of sec *x*.**

  – **Powers of sec *x* may require integration by parts, as shown in the following example.**

STRATEGY C

- If the powers of both sine and cosine are even, use the half-angle identities

$$\sin^2 x = \tfrac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \tfrac{1}{2}(1 + \cos 2x)$$

Sometimes, it is helpful to use the identity

$$\sin x \cos x = \tfrac{1}{2}\sin 2x$$

For other cases, the guidelines are not as clear-cut.

We may need to use:

Identities

Integration by parts

A little ingenuity

Example

Show that

$$\int \tan x \, dx = \ln |\sec x| + C$$

(we prove it before in section 5.5 page 416)

Example

Show that

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

Integrals of the form $\int \cot^m x \csc^n x \, dx$ can be found by similar methods.

We have to make use of the identity :

$$1 + \cot^2 x = \csc^2 x$$

OTHER INTEGRALS

- In order to evaluate the integral, use the corresponding identity.

	Integral	Identity
a	$\int \sin^m x \cos^n x \, dx$	$\sin A \cos B = \frac{1}{2}\left[\sin(A-B)+\sin(A+B)\right]$
b	$\int \sin^m x \sin^n x \, dx$	$\sin A \sin B = \frac{1}{2}\left[\cos(A-B)-\cos(A+B)\right]$
c	$\int \cos^m x \cos^n x \, dx$	$\cos A \cos B = \frac{1}{2}\left[\cos(A-B)+\cos(A+B)\right]$

Example

- **Evaluate** $\int \sin 4x \cos 5x \, dx$

	Integral	Identity
a	$\int \sin^m x \cos^n x \, dx$	$\sin A \cos B = \frac{1}{2}\left[\sin(A-B)+\sin(A+B)\right]$
b	$\int \sin^m x \sin^n x \, dx$	$\sin A \sin B = \frac{1}{2}\left[\cos(A-B)-\cos(A+B)\right]$
c	$\int \cos^m x \cos^n x \, dx$	$\cos A \cos B = \frac{1}{2}\left[\cos(A-B)+\cos(A+B)\right]$

Example*

- **Evaluate** $\int \{[\cos x+\sin x] / \sin(2x)\} dx$



Example

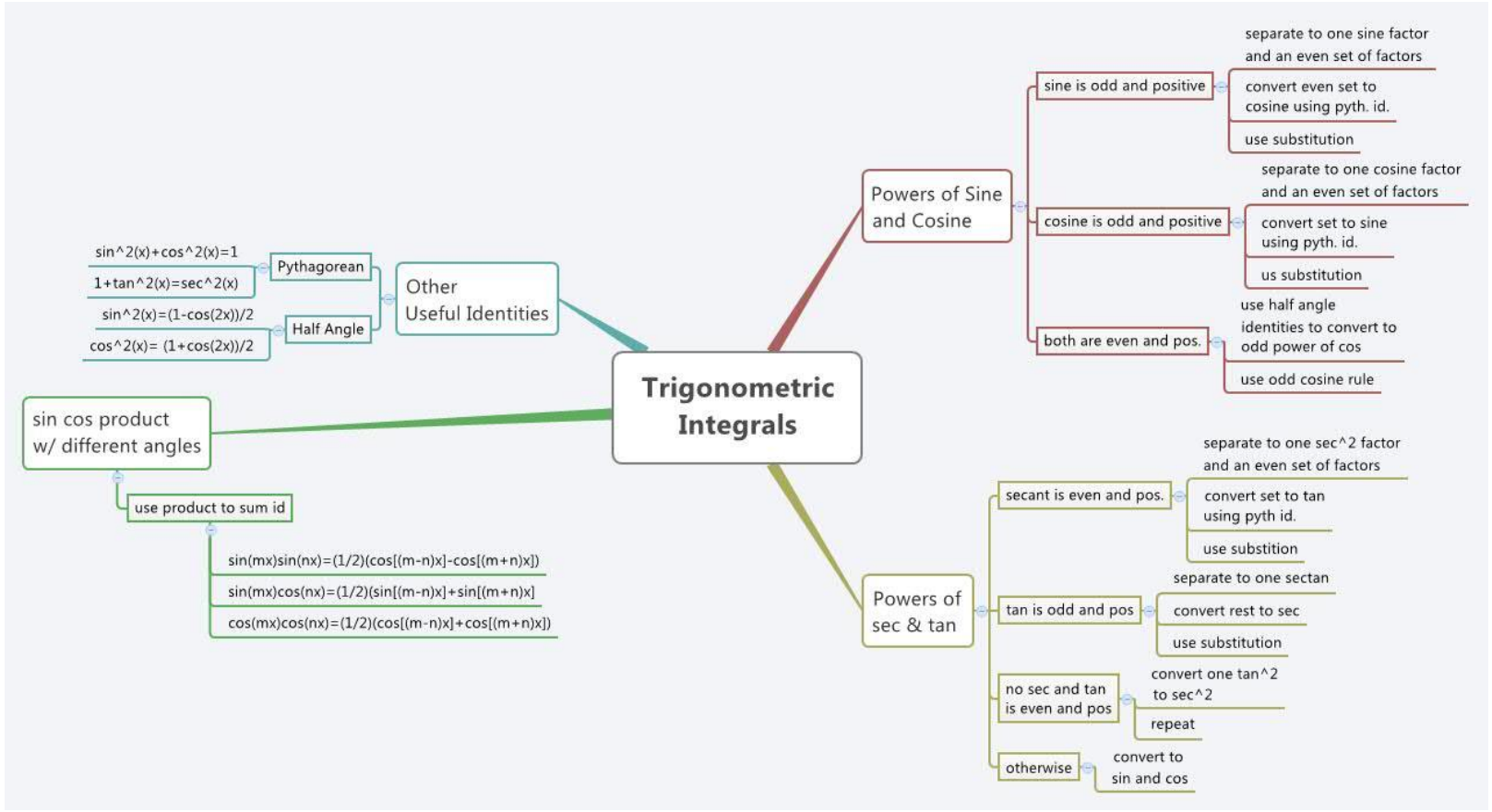
- Find $\int \cos^2 x \tan^3 x \, dx$



Example(HW):

Find

- $\int \tan^5 x \sec^7 x \, dx$
- $\int \sec^3 x \, dx$
- $\int \sin^5 x \, dx$
- $\int \cos^2 x \tan^3 x \, dx$
- $\int \csc x \, dx$


Trigonometric Integrals
Powers of Sine and Cosine
sine is odd and positive
separate to one sine factor and an even set of factors
convert even set to cosine using pyth. id.
use substitution
cosine is odd and positive
separate to one cosine factor and an even set of factors
convert set to sine using pyth. id.
us substitution
both are even and pos.
use half angle identities to convert to odd power of cos
use odd cosine rule
Powers of sec & tan
secant is even and pos.
separate to one sec^2 factor and an even set of factors
convert set to tan using pyth id.
use substition
tan is odd and pos
separate to one sectan
convert rest to sec
use substitution
no sec and tan is even and pos
convert one tan^2 to sec^2
repeat
otherwise
convert to sin and cos
Other Useful Identities
Pythagorean
sin^2(x)+cos^2(x)=1
1+tan^2(x)=sec^2(x)
Half Angle
sin^2(x)=(1-cos(2x))/2
cos^2(x)= (1+cos(2x))/2
sin cos product w/ different angles
use product to sum id
sin(mx)sin(nx)=(1/2)(cos[(m-n)x]-cos[(m+n)x])
sin(mx)cos(nx)=(1/2)(sin[(m-n)x]+sin[(m+n)x]
cos(mx)cos(nx)=(1/2)(cos[(m-n)x]+cos[(m+n)x])